

▼ **Figure 21.3 National Center for Biotechnology Information (NCBI) website.** Some results are shown from a search for regions of proteins similar to an amino acid sequence in a muskmelon protein.

The figure displays a screenshot of the NCBI website interface. At the top, a window titled "WD40 - Sequence Alignment Viewer" shows a sequence alignment. A "Query" sequence is at the top, followed by several other sequences from different organisms, including Cow [transducin], Mustard weed [transducin], Corn [GNB protein], Human [PAFA protein], Nematode [unknown protein #1], Nematode [unknown protein #2], and Fission yeast [FWDR protein]. The alignment shows regions of similarity highlighted in yellow. Below the alignment, a window titled "WD40 - Cn3D 4.1" displays a 3D ribbon model of a protein domain. The model is purple, with some segments highlighted in yellow. To the left of the 3D model, a window titled "CDD Descriptive Items" provides information about the WD40 domain, including its function and structure. Numbered callouts (1-6) point to specific features in the image.

1 In this window, a partial amino acid sequence from an unknown muskmelon protein ("Query") is aligned with sequences from other proteins that the program found to be similar. Each sequence represents a domain called WD40.

2 Four hallmarks of the WD40 domain are highlighted in yellow. (Sequence similarity is based on chemical aspects of the amino acids, so the amino acids in each hallmark region are not always identical.)

3 The Cn3D ("See in 3D") program displays 3D models of domains, such as this ribbon model of cow transducin (the protein highlighted in purple in the Sequence Alignment Viewer). This protein is the only one of those shown for which a structure has been determined. The sequence similarity of the other proteins to cow transducin suggests that their structures are likely to be similar.

4 Cow transducin contains seven WD40 domains, one of which is highlighted here in gray.

5 The yellow segments correspond to the WD40 hallmarks highlighted in yellow in the window above.

6 This window displays information about the WD40 domain from the Conserved Domain Database (CDD), which can find and describe similar domains in related proteins.

Identifying Protein-Coding Genes and Understanding Their Functions

Using available DNA sequences, geneticists can study genes directly, rather than taking the classical genetic approach, which requires determining the function of an unknown gene from the phenotype. But this more recent approach poses a new challenge: What does the gene actually do? Given a long DNA sequence from a database like GenBank, scientists aim to identify all protein-coding genes in the sequence and ultimately their functions. This process, called **gene annotation**, uses three lines of evidence to identify a gene.

First, computers are utilized in a search for patterns that indicate the presence of genes. The usual approach is to use software to scan the stored sequences for those that represent transcriptional and translational start and stop signals, RNA-splicing sites, and other telltale signs of protein-coding genes, such as promoter sequences. The software also looks for certain short sequences that specify known mRNAs. Thousands of such sequences, called *expressed sequence tags*, or *ESTs*, have been collected from cDNA sequences and are cataloged in computer databases. This type of analysis identifies sequences that may turn out to be previously unknown protein-coding genes.

Although the identities of about half of the human genes were known before the Human Genome Project began, the other genes, previously unknown, were revealed by DNA sequence analysis. Once such suspected genes are identified, the second step is to obtain clues about their identities and functions. Software is used to compare their sequences with those of known genes from other organisms. Due to redundancy in the genetic code, the DNA sequence itself may vary more among species than the protein sequence does. Thus, scientists interested in proteins often compare the predicted amino acid sequence of a protein to that of other proteins. The final step is to confirm the identities of these genes using RNA-seq (see Figure 20.12) or some other method to show that the relevant RNA is actually expressed from the proposed gene.

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Sometimes a newly identified sequence will match, at least partially, the sequence of a gene or protein in another species whose function is well known. For example, a plant researcher working on signaling pathways in the muskmelon would be excited to see that a partial amino acid sequence from a gene she had identified matched sequences in other